



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

DHL AUTOMATION CHALLENGE 2026: DHL LOGISENSE

WEB TECHNOLOGY

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1. Introduction

DHL LogiSense is an AI-assisted logistics knowledge management system. It was developed to help DHL operations team convert unstructured operational source material into structured Standard Operating Procedures (SOPs) and knowledge articles. The project combines a deployed web application with a UiPath RPA workflow that monitors a OneDrive folder, extracts text from different file formats and sends the result to the LogiSense API.

The purpose of the system is to reduce the manual effort required to turn daily logistics issues into reusable operational knowledge. Instead of relying on scattered messages, screenshots, email threads, PDF documents, and handwritten notes, LogiSense centralizes the material, stores the original source evidence, and generates reviewable documents for authorized users.

The final product uses Next.js, Payload CMS, PostgreSQL, Auth0 authentication, OpenAI-based transformation, UiPath automation, and a dashboard interface. Human review remains part of the process: the AI drafts the document, but reviewers can compare the generated output with the original raw source text before relying on it.

2. Scenario Description

The scenario focuses on a logistics operations environment where new issues are continuously created by staff, customers, drivers, gateway teams, billing teams, compliance teams, and control tower personnel. These issues may arrive in many forms, such as a Teams conversation about an invalid postcode, an email thread about a missing commercial invoice, a screenshot showing a printer error, a PDF, a Word file, a plain text note, or an Outlook .msg file.

In the original manual scenario, a staff member would need to read the source material, interpret the problem, summarize it, decide whether it should become an SOP or an article, write the content, and store it somewhere for future reference. This is slow and inconsistent because each source format requires different handling and each reviewer may write the final guidance differently.

The LogiSense scenario automates the intake path. A file is placed in the OneDrive /Issues folder. UiPath detects the file, moves it to a /Processed folder, extracts text according to file type, prepares a summary, and posts the structured payload to the deployed LogiSense application. The web application stores the raw input and uses AI to create a category, title, document type, and formatted content.

- Input source: Operational files placed in the monitored OneDrive /Issues folder.
- Automation layer: UiPath workflow reads, do OCR, parses and submits the file content.
- Backend layer: LogiSense API stores raw inputs and triggers document generation.
- Knowledge layer: Generated SOPs and articles are shown in a secure review dashboard.
- Review layer: T1 users can inspect documents, while T2 users can edit or delete generated knowledge.

3. API & JSON Structure

The main integration point between UiPath and the deployed LogiSense application is the ingestion API. The RPA workflow sends a JSON request to the deployed endpoint after it extracts raw content and summary information from the uploaded file.

Table 3-1 Main Ingestion Endpoint

Item	Value	Explanation
Method	POST	Creates a new raw input record and starts AI generation.
Endpoint	https://dhl-logi-sense.vercel.app/api/ingest	Receives the payload from UiPath.
Content Type	application/json	The request body is sent as JSON.
Duplicate key	sourceId	Prevents the same OneDrive file from being processed twice (over the cloud issue).

Request JSON:

```
{
  "rawText": "Full extracted text or OCR output from the source file",
  "summary": "Short summary or file-based title used by the AI",
  "sourceType": "txt | docx | pdf | msg | jpg | png | tif | other extension",
  "sourceId": "Unique OneDrive file identifier"
}
```

The backend normalizes the sourceType, checks whether the sourceId already exists, creates a RawInput record, and then runs AI generation step in the background. The rawText

field is the most important field because it contains the operational evidence used to generate the SOP or article.

Successful Response JSON:

```
{
  "success": true,
  "docId": 123
}
```

Duplicate Response JSON:

```
{
  "success": true,
  "message": "Already processed"
}
```

AI Output JSON:

```
{
  "category": "Errors",
  "documentType": "sop",
  "title": "Invoice Mismatch Resolution Procedure",
  "content": "<h1>...</h1><p>...</p><ul><li>...</li></ul>"
}
```

The AI output is not returned directly to UiPath. It is used internally by the backend to create either an SOP record or an Article record in PayloadCMS. The content field is HTML so that it can be loaded onto the TipTap editor on the document review page.

MSG Parsing Endpoint:

```
{
  "rawText": "Subject, sender, recipients, date, attachments,
and email body",
  "summary": "Email subject or filename"
}
```

The workflow also uses a helper endpoint for Outlook message files. For .msg files, UiPath downloads the file locally and posts it as multipart form data to /api/parse-msg. The

endpoint extracts subject, sender, recipients, date, attachments, and body text, then returns rawText and summary. UiPath then sends those values to /api/ingest using the normal JSON structure.

4. User Interface Screenshots

The screenshots below were taken from the deployed site at <https://dhl-logi-sense.vercel.app/>. The account is shown as a T1 reviewer, meaning it can inspect generated SOPs/articles and compare source evidence, but it does not expose edit and delete controls reserved for T2 editors.

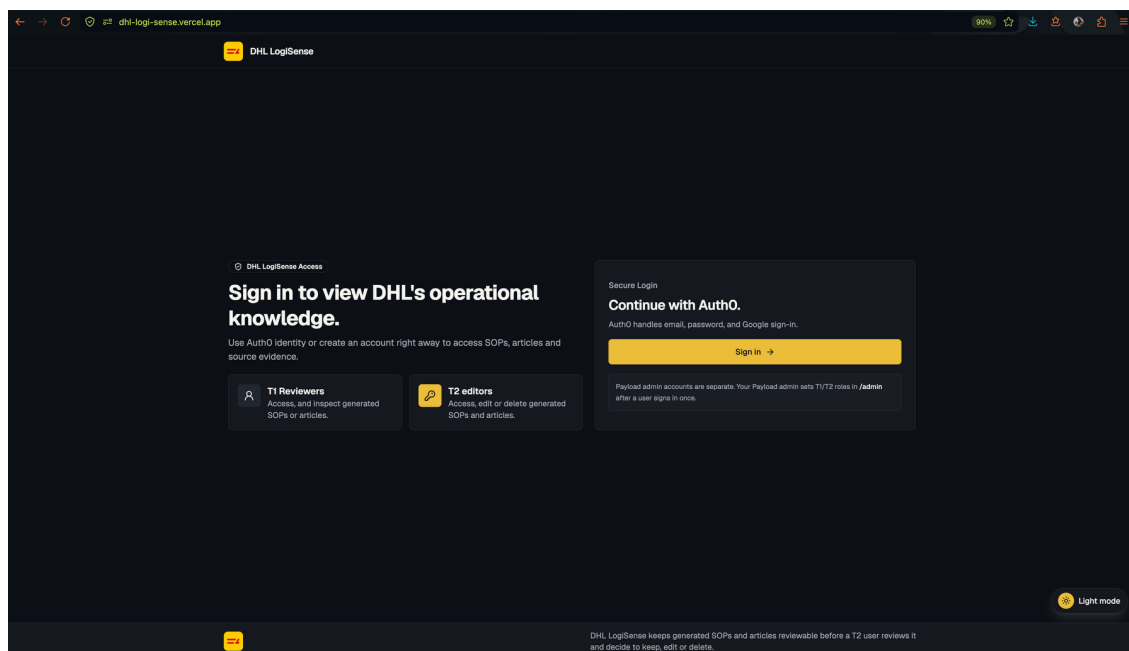


Figure 1 LogiSense Login Page

The login screen explains that DHL LogiSense uses Auth0 for identity. It also introduces the two user roles: T1 reviewers for inspection and T2 editors for editing or deleting generated knowledge.

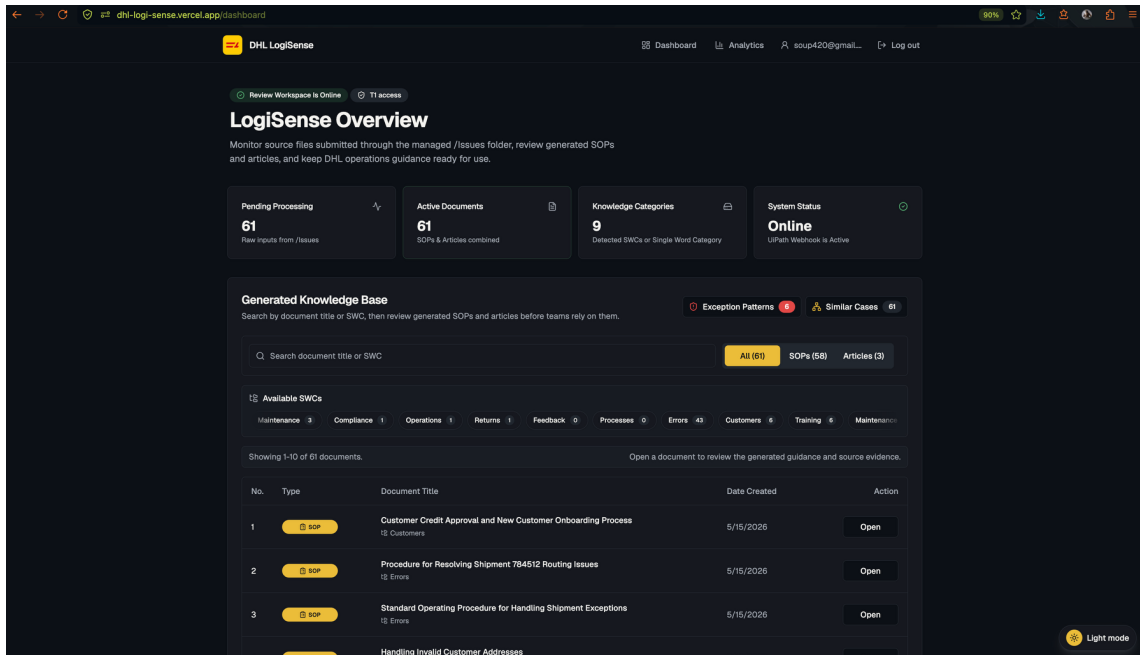


Figure 2 Dashboard Overview and Generated Knowledge Base

After login, the dashboard shows the operation status of the system. In the captured deployment, there are 61 raw inputs, 61 active generated documents, 9 detected SWCs (single word category) and the UiPath webhook is online. The table lists generated SOPs and articles with search, document type filtering, SWC filtering and Open actions.

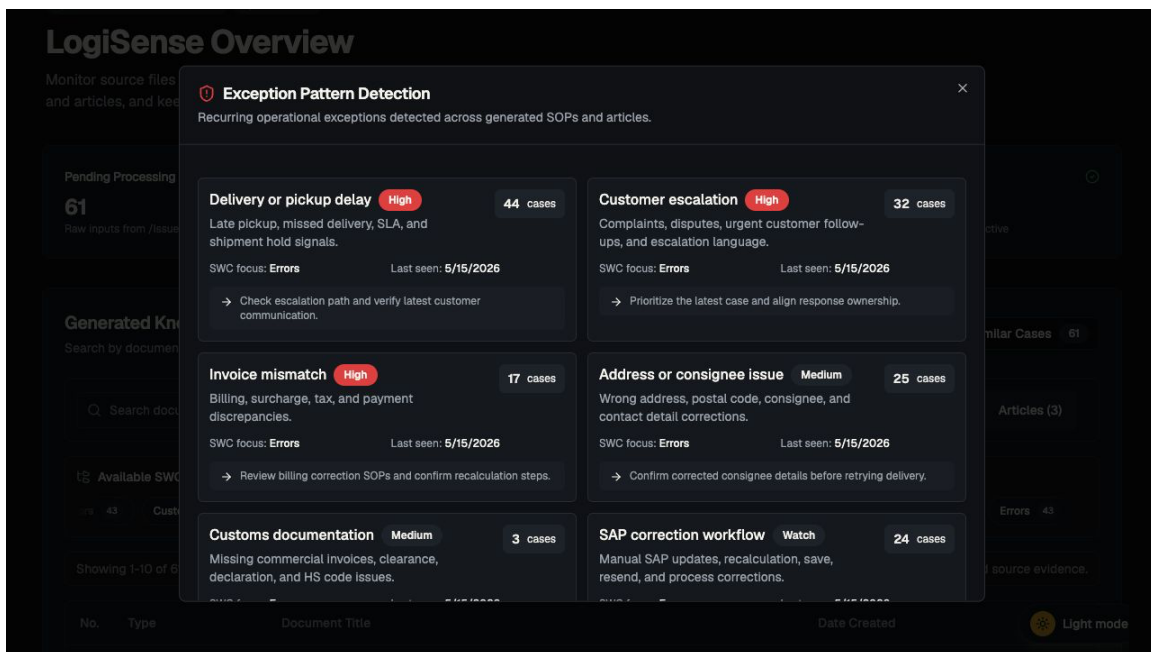


Figure 3 Exception Pattern Detection

The exception pattern detection modal groups recurring operational problems found across generated content. Examples shown include delivery or pickup delay, customer escalation, invoice mismatch, address or consignees' issues, customs documentations and SAP correction workflow. Each pattern includes severity, case count, SWC focus, last-seen date and recommended next action.

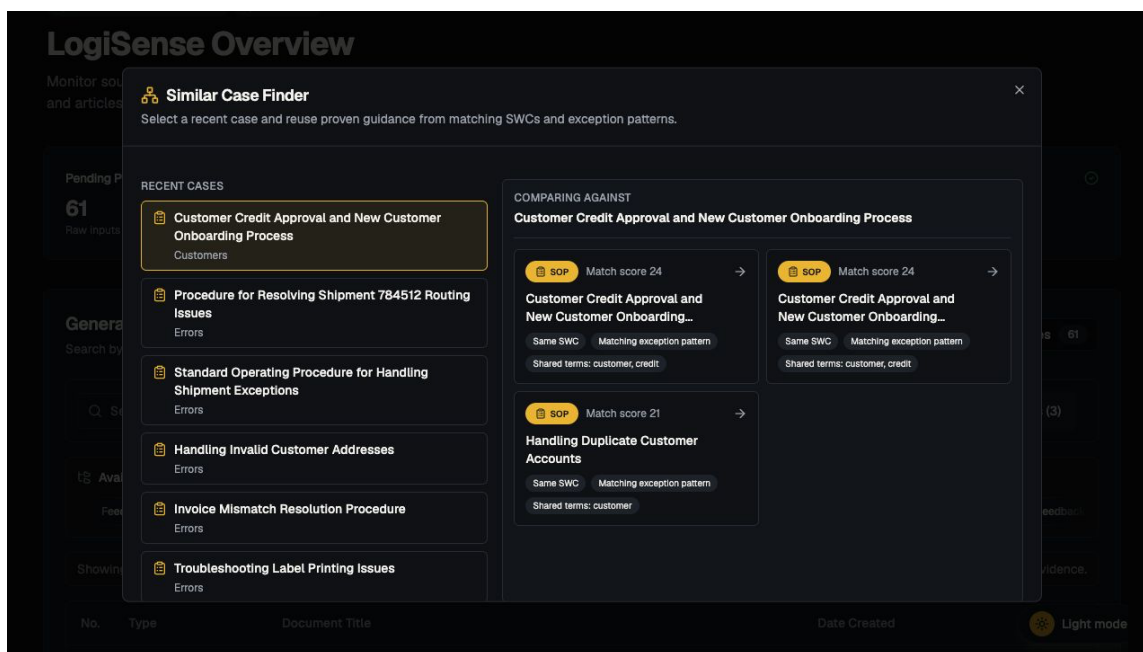


Figure 4 Similar Case Finder

The similar case finder lets reviewers compare a selected case with related existing documents. It uses signals such as same SWC, matching exception pattern, shared title terms and source types. This helps terms reuse previous guidance rather than starting from scratch for every similar issue.

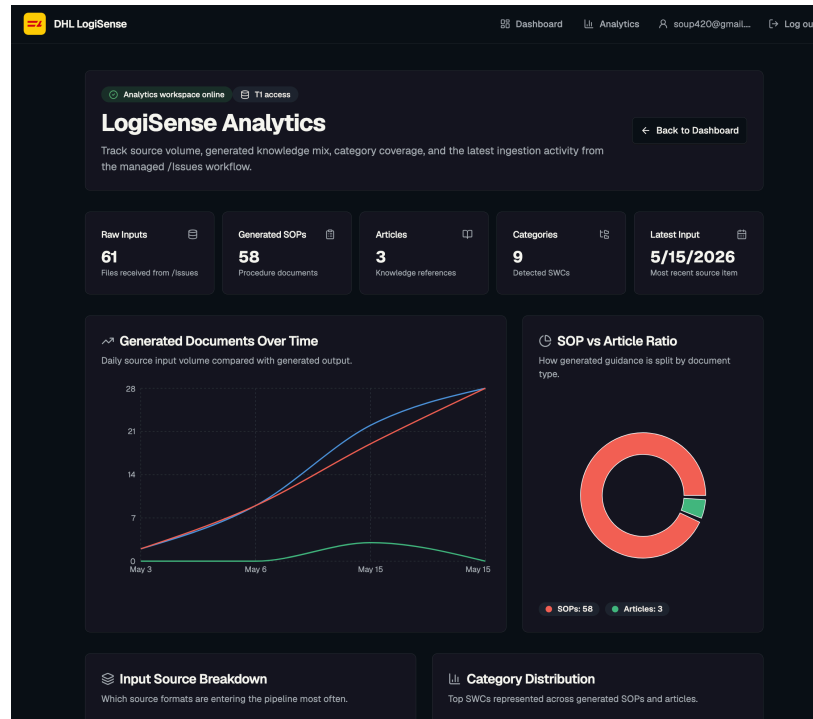


Figure 5 Analytics Dashboard

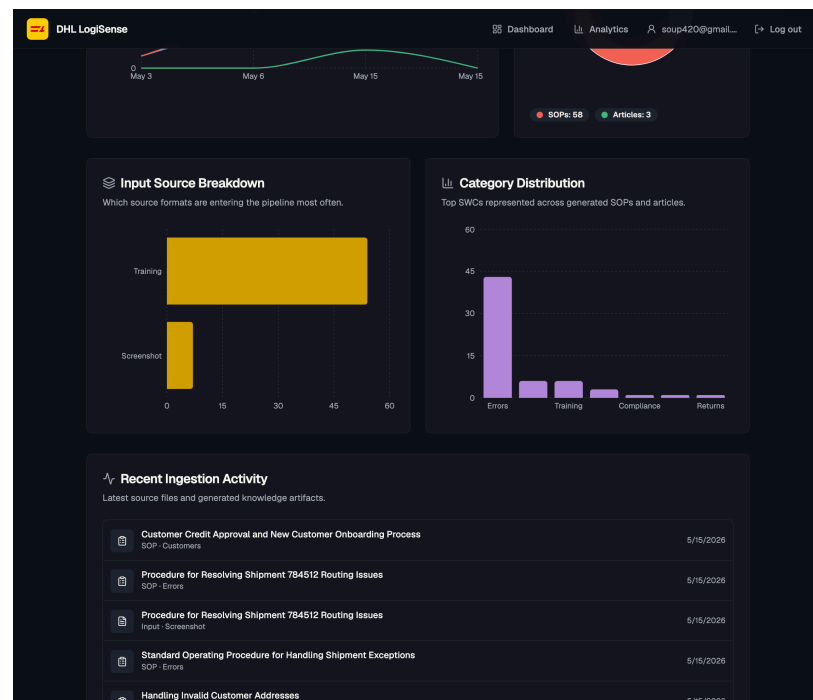


Figure 6 Continuation of Analytics Dashboard

The analytics page summarizes source volume and generated knowledge output. It shows count for raw inputs, generated SOPs, articles, categories and latest input date. It also

visualizes generated documents over time, SOP versus article ratio, source type breakdown, category distribution and recent ingestion activity.

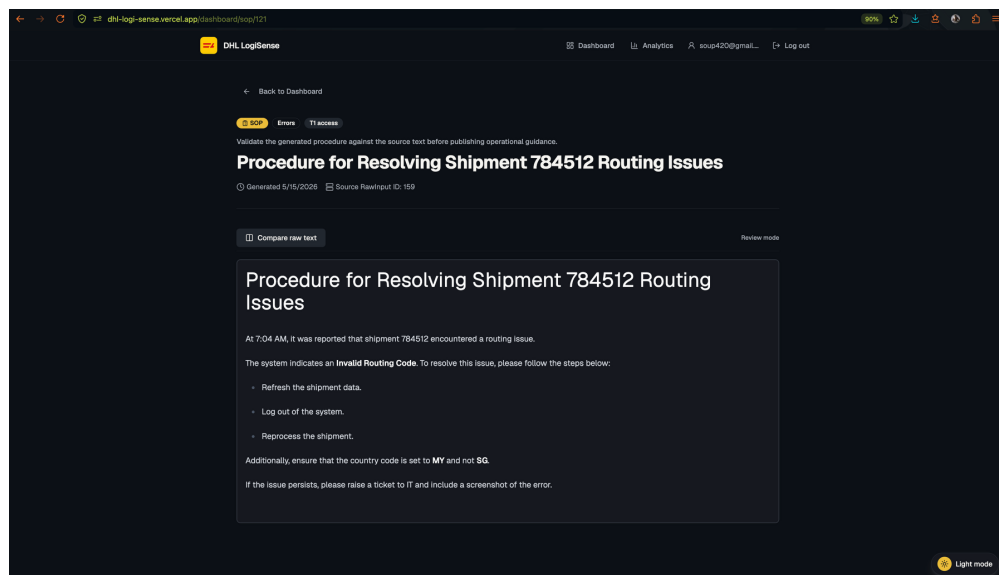


Figure 7 SOP Document Review Page

The document page displays the selected SOP with metadata such as category, role, generation date and source RawInput ID. In T1 mode, the content is review-only. The reviewer can read the generated SOP and decide whether it is suitable for operational use.

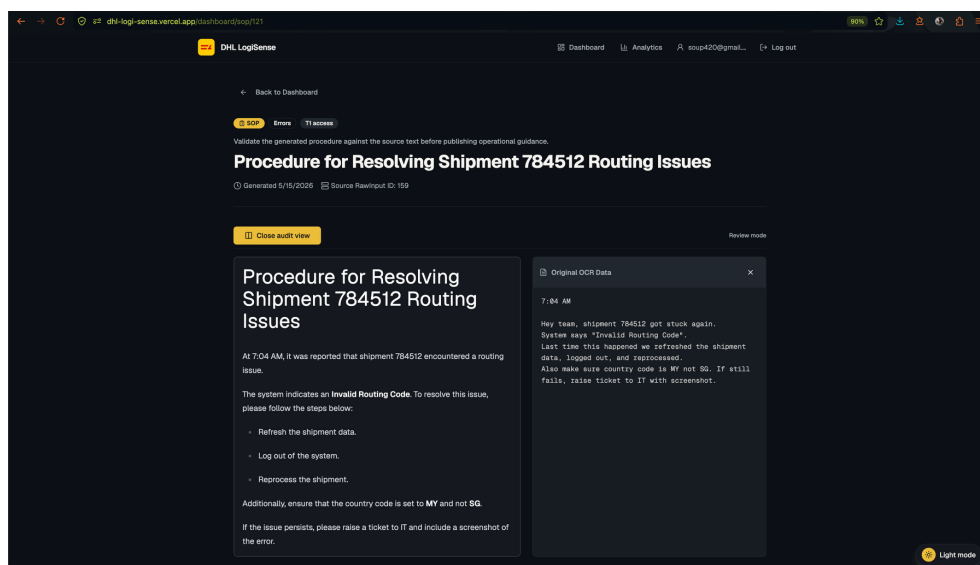


Figure 8 Raw Source Comparison View

The compare raw text feature opens the original OCR/source text beside the generated SOP. This is a critical governance future because it allows reviewers to verify whether the generated guidance accurately reflects the source material before it is used by operations teams.

5. RPA Workflow & Automation Explanation

It implements the automation bridge between OneDrive source files and the deployed LogiSense API. The workflow starts when a new file is created in the OneDrive /Issues folder, processes the file according to its file extension, then sends the extracted content to the LogiSense ingestion endpoint.

Variable	Purpose	How it is Used
file	New OneDrive item from the event listener trigger (file creation).	Captures the file created in the /Issues folder.
fileOrFolder	Moved OneDrive item (to /Processed).	Represents the file after it is moved to /Processed.
varRawContent	Extracted text.	Sent as rawText to /api/ingest.
varSummary	Summary or filename.	Sent as summary to api/ingest.
documentData	OCR/generative extraction result.	Used for image or the default branch extraction.
msgDownloadFolder	Temporary folder for .msg files.	Stores downloaded Outlook messages before parsing.
msgLocalPath	Local path to download .msg file.	Passed to /api/parse-msg as multipart file data.
parseMsgResponse	HTTP result from .msg parser.	Parsed to fill varRawContent and varSummary.

The RPA workflow acts as the intake automation for the whole LogiSense system. It removes the need for a user to manually upload data into the web app. Staff only need to place a file into the OneDrive /Issues folder. UiPath handles the detection, extraction, file movement, parsing, and API submission. This turns OneDrive into a practical operational inbox for logistics knowledge generation.

The use of different extraction branches is important because logistics evidence does not arrive in one standard format. Text files and Word documents can be read directly, PDFs

require OCR, images require document extraction, and Outlook .msg files require a special parser. By normalizing all of these into the same JSON structure, the backend can treat every source consistently.

Step-by-step automation:

1. File Created trigger monitors the OneDrive /Issues folder. When a new file appears, the workflow receives its metadata as the file variable.
2. Move File or Folder moves the new item from /Issues to the /Processed folder. This prevents the same item from staying in the intake folder and makes the automation flow cleaner.
3. The workflow logs the detected file extension using `fileOrFolder.Extension.TrimStart(".").Trim().ToLowerInvariant()`.
4. A Switch activity routes the file based on extension. Supported explicit branches include txt, docx, pdf, and msg. Other files go to the default image/OCR branch.
5. For image or default files, UiPath uses Extract Document Data with a generative extractor. The prompt asks it to summarize the document into a clear SOP title. `documentData.DocumentMetadata.Text` becomes `varRawContent`, and `documentData.Data.Summary.Value` becomes `varSummary`.
6. For txt files, UiPath reads the text directly using Read Text File and uses the filename as the summary.
7. For docx files, UiPath uses the Word Read Text activity and uses the filename as the summary.
8. For pdf files, UiPath uses Extract PDF Text with OCR enabled and uses the filename as the summary.
9. For msg files, UiPath creates a temporary folder, downloads the message, posts it to <https://dhl-logi-sense.vercel.app/api/parse-msg>, then parses the JSON response to set `varRawContent` and `varSummary`.
10. The workflow logs the raw text and summary for debugging.
11. Finally, UiPath sends an HTTP POST request to <https://dhl-logi-sense.vercel.app/api/ingest> with `rawText`, `summary`, `sourceType`, and `sourceId`.

Final JSON Payload Built by UiPath:

```
{
  "rawText": varRawContent,
  "summary": varSummary,
  "sourceType":
    fileOrFolder.Extension.TrimStart(".").Trim().ToLowerInvariant(),
```

```
"sourceId": fileOrFolder.Id  
}
```

6. Conclusion

DHL LogiSense provides an end-to-end solution for converting unstructured logistics source material into reviewable SOPs and knowledge articles. The deployed interface allows users to sign in securely, view operational metrics, search generated documents, identify recurring exception patterns, find similar cases, analyze ingestion trends, and compare generated guidance with original source text.

The UiPath workflow is a key part of the system because it automates the intake of files from OneDrive and submits a clean JSON payload to the LogiSense API. This connects real operational files to the web application without requiring manual copying and pasting. The API structure is simple but effective: `rawText`, `summary`, `sourceType`, and `sourceId` are enough for the backend to store evidence and generate knowledge.

Overall, the project demonstrates how RPA and AI can work together in a logistics environment. UiPath handles repetitive file monitoring and extraction tasks, while LogiSense handles storage, AI transformation, review, analytics, and governance. The result is a practical system that improves knowledge reuse while still keeping humans responsible for final review and operational trust.